Module 7

Using Data Types and Units

ITCL112 - INTRODUCTION TO COMPUTING W/ LAB

INTRODUCTION

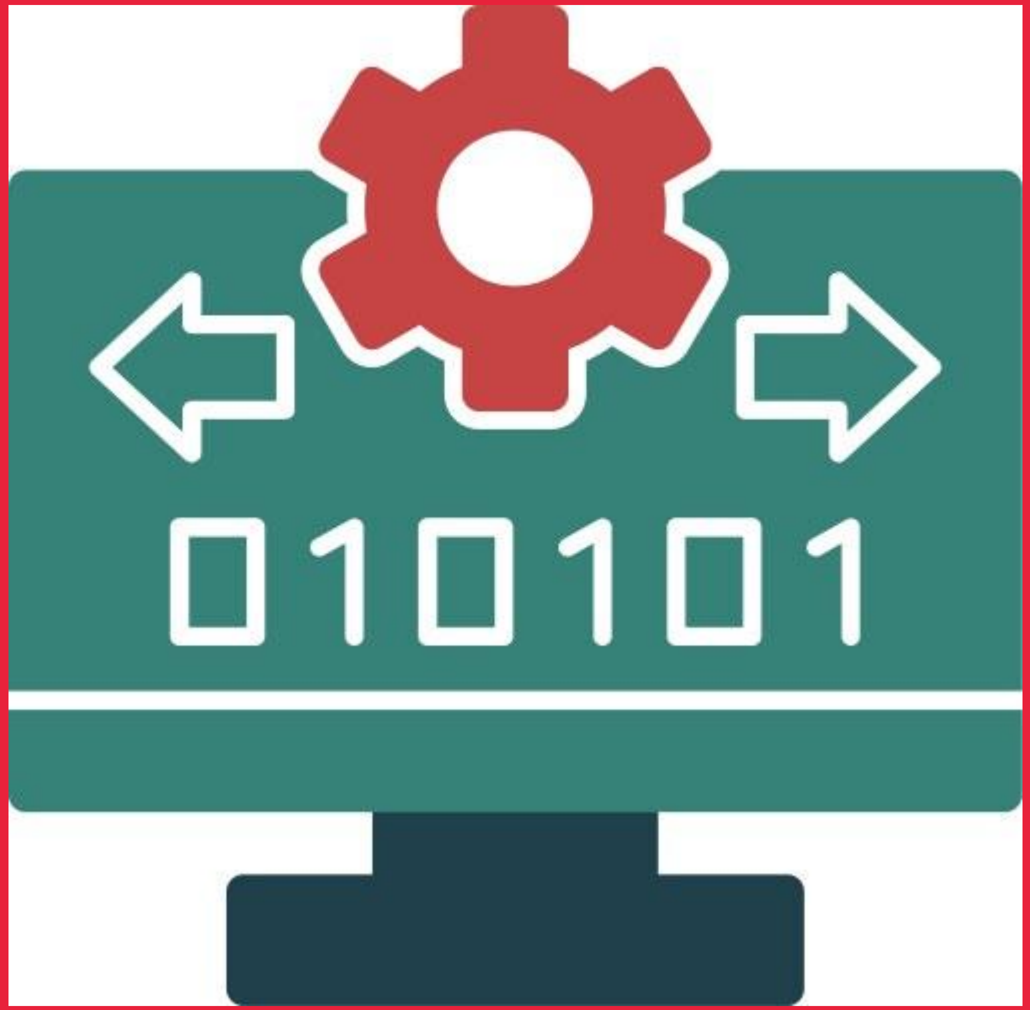
Computers are digital. This means that they operate on numbers, be it photos, music, or text documents; everything within a computer is represented as a number, and all operations are some form of arithmetic.

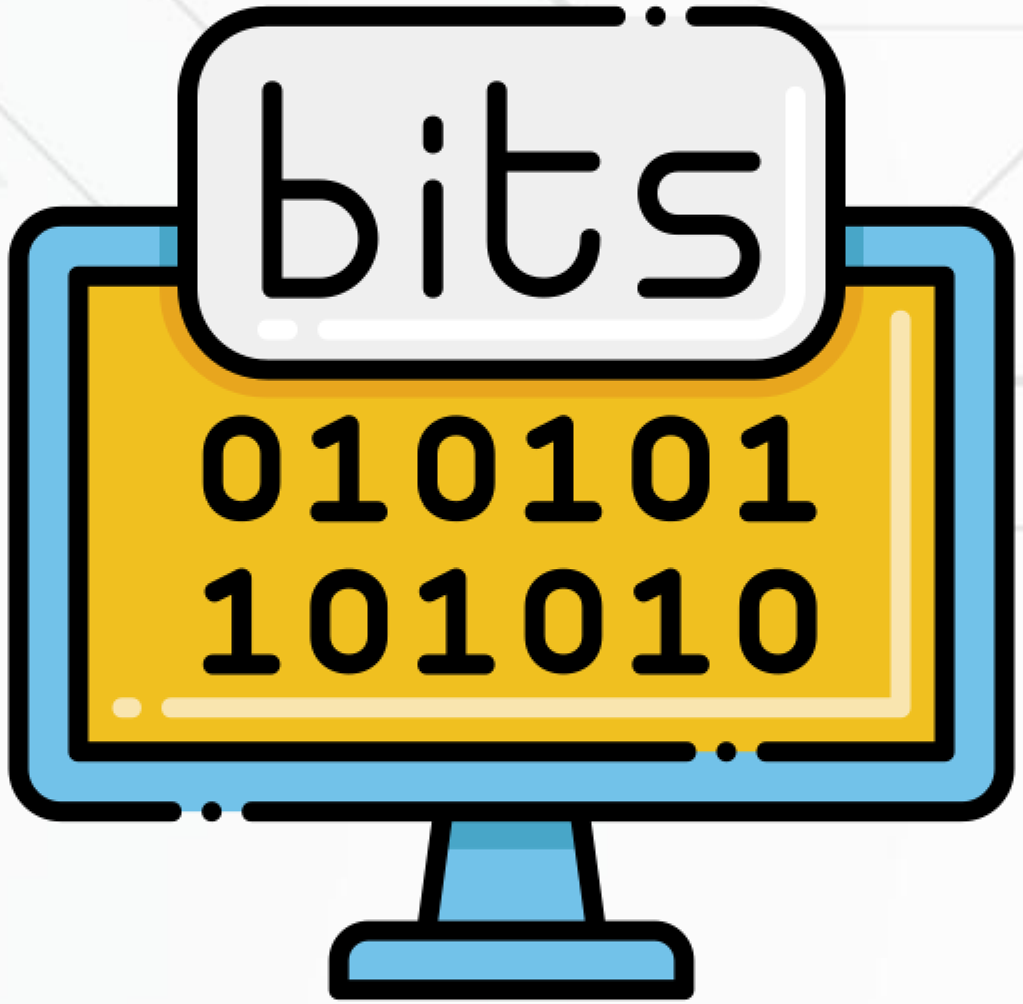
MODULE 6

1. Units of Measures in Computing
2. Throughput Units and Processing Speed Units
3. Data Types: Integer, Floating-point numbers, Boolean, Characters and Strings
4. Data Representation of ASCII and Unicode

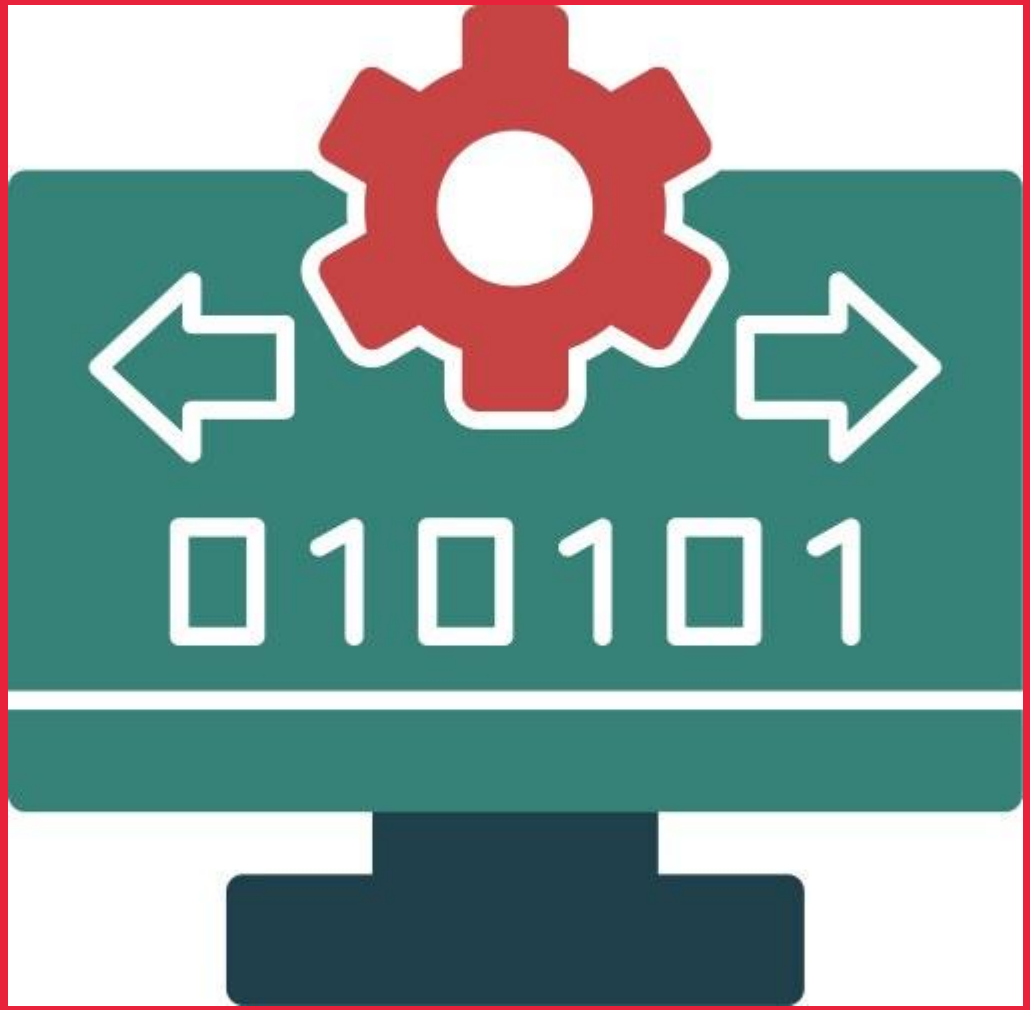
Bits and Bytes

It all starts with a bit (the [binary] digit), which makes up the smallest unit of data for computers. Being a binary value, this unit can only have one of two values: 0 or 1, usually to represent if an electronic signal is on or off.

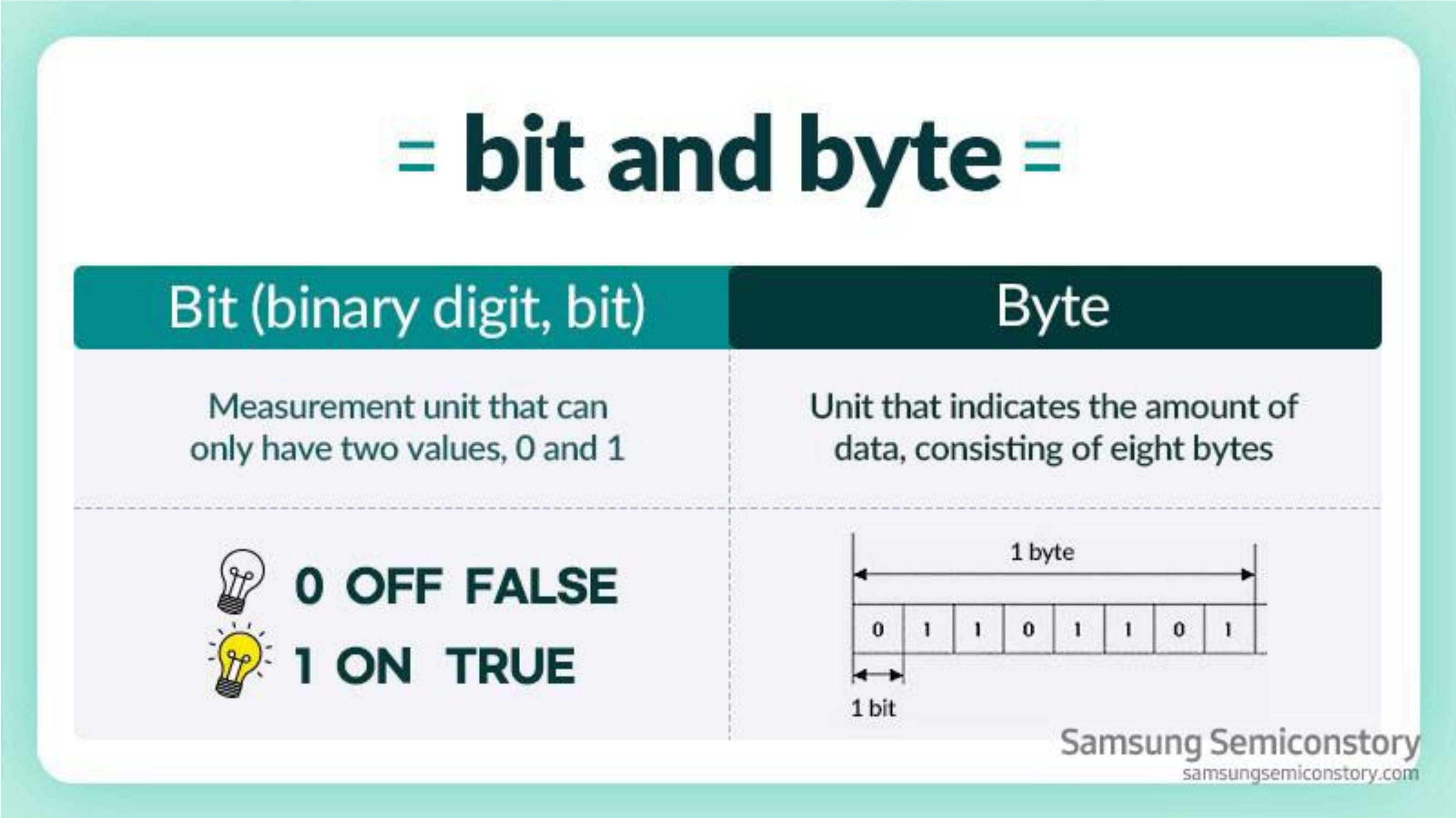




Bits and Bytes



All data on the computer is represented in this binary notation. For example, the number 228 can be represented as 11100100 in binary, and a lowercase j can be represented as 01101010 in the ASCII encoding standard. These groupings of 8 bits are so common that it has a special name, a byte



Bytes

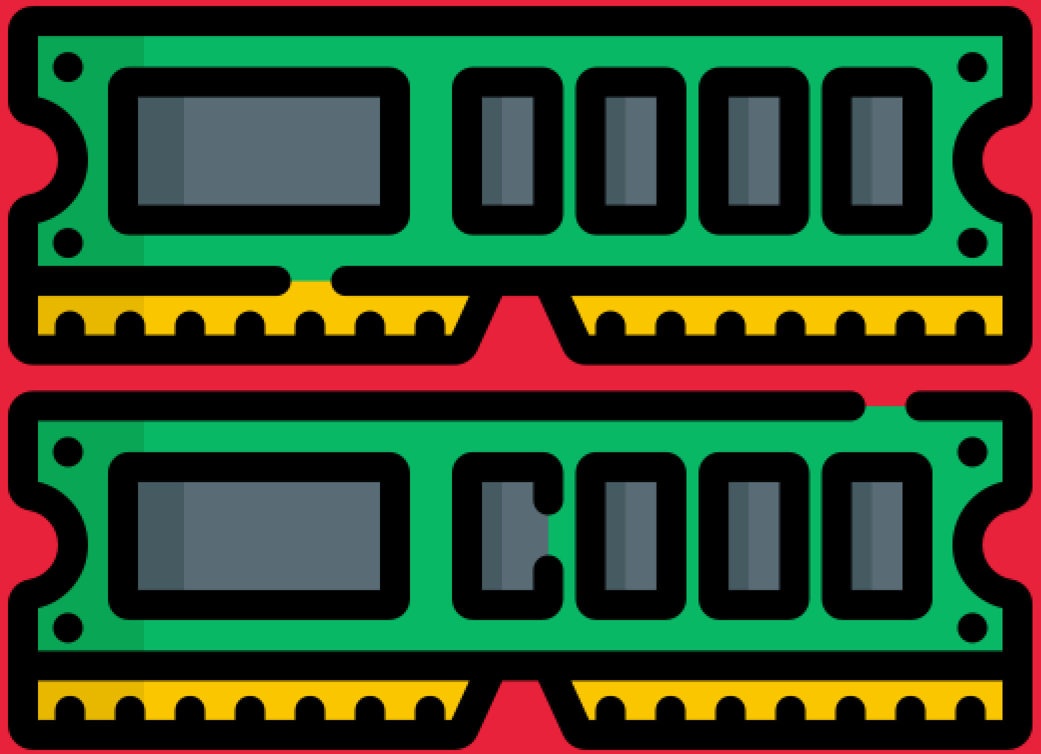
Memory Capacity Conversion Chart

Term (Abbreviation)	Approximate Size
Byte (B)	8 bits
Kilobyte (KB)	1024 bytes / 10^3 bytes
Megabyte (MB)	1024 KB / 10^6 bytes
Gigabyte (GB)	1024 MB / 10^9 bytes
Terabyte (TB)	1024 GB / 10^{12} bytes
Petabyte (PB)	1024 TB / 10^{15} bytes
Exabyte (EB)	1024 PB / 10^{18} bytes
Zettabyte (ZB)	1024 EB / 10^{21} bytes
Yottabyte (YB)	1024 ZB / 10^{24} bytes

Measuring Storage Performance

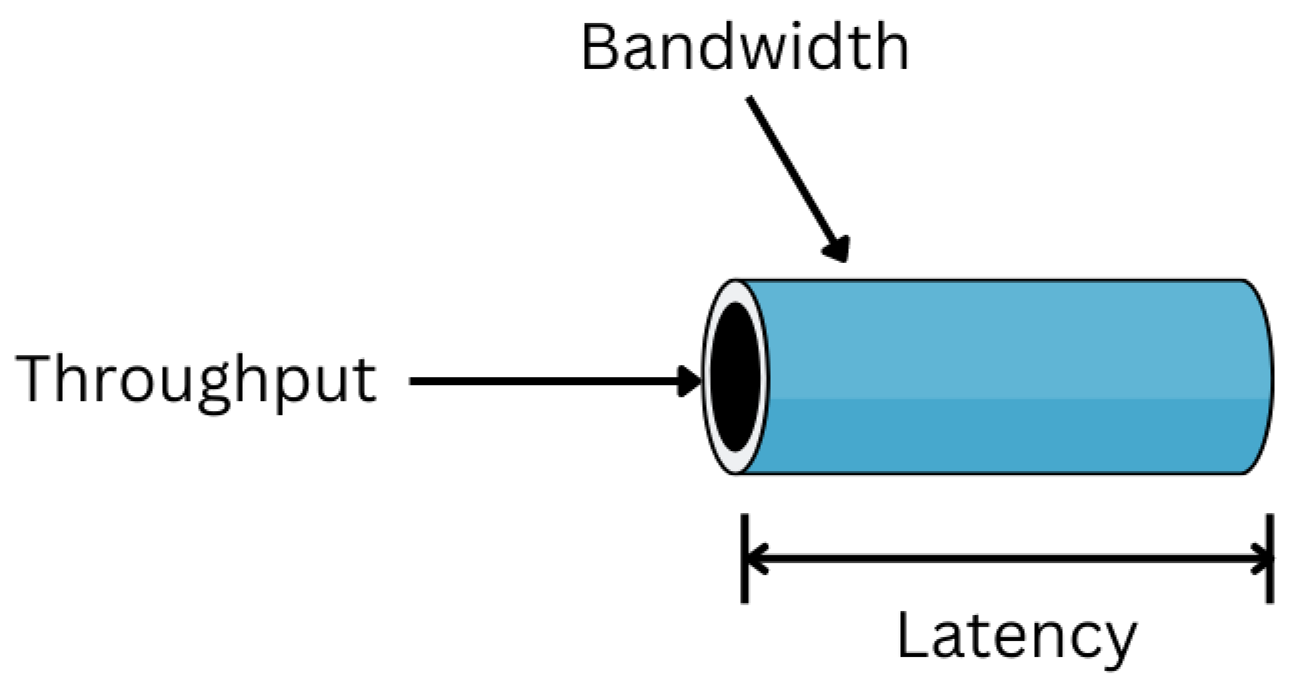
- ✓ If hard drives only had a speed of 1 megabyte per second (MBps), then the example song above would take 5 seconds to load before playing!
- ✓ Modern drives are much faster than that. The storage of most devices today are solid state drives. For portable devices such as phones, they are usually able to read files at about 1500 MBps and write new files at 500 MBps. Desktop devices are able to reach read and write speeds of over 5000 MBps!

Throughput Units and Processing Speed Unit

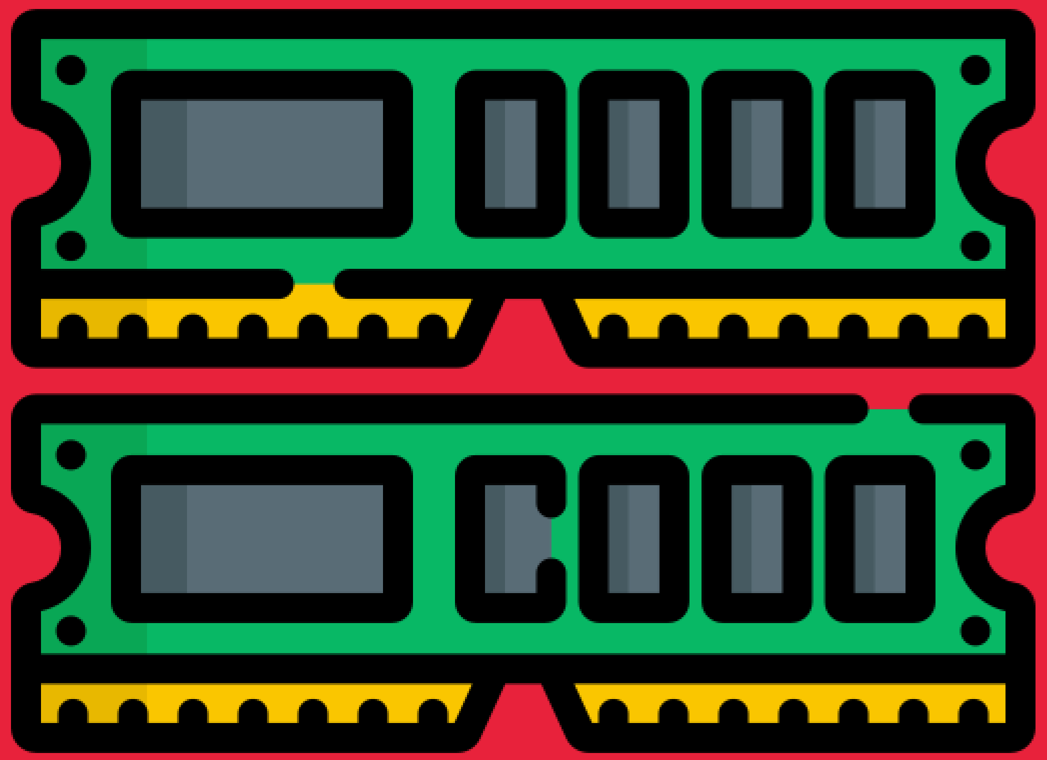


Throughput is the total amount of data that can be transferred during a given amount of time.

Latency is the amount of delay before that transfer of data begins

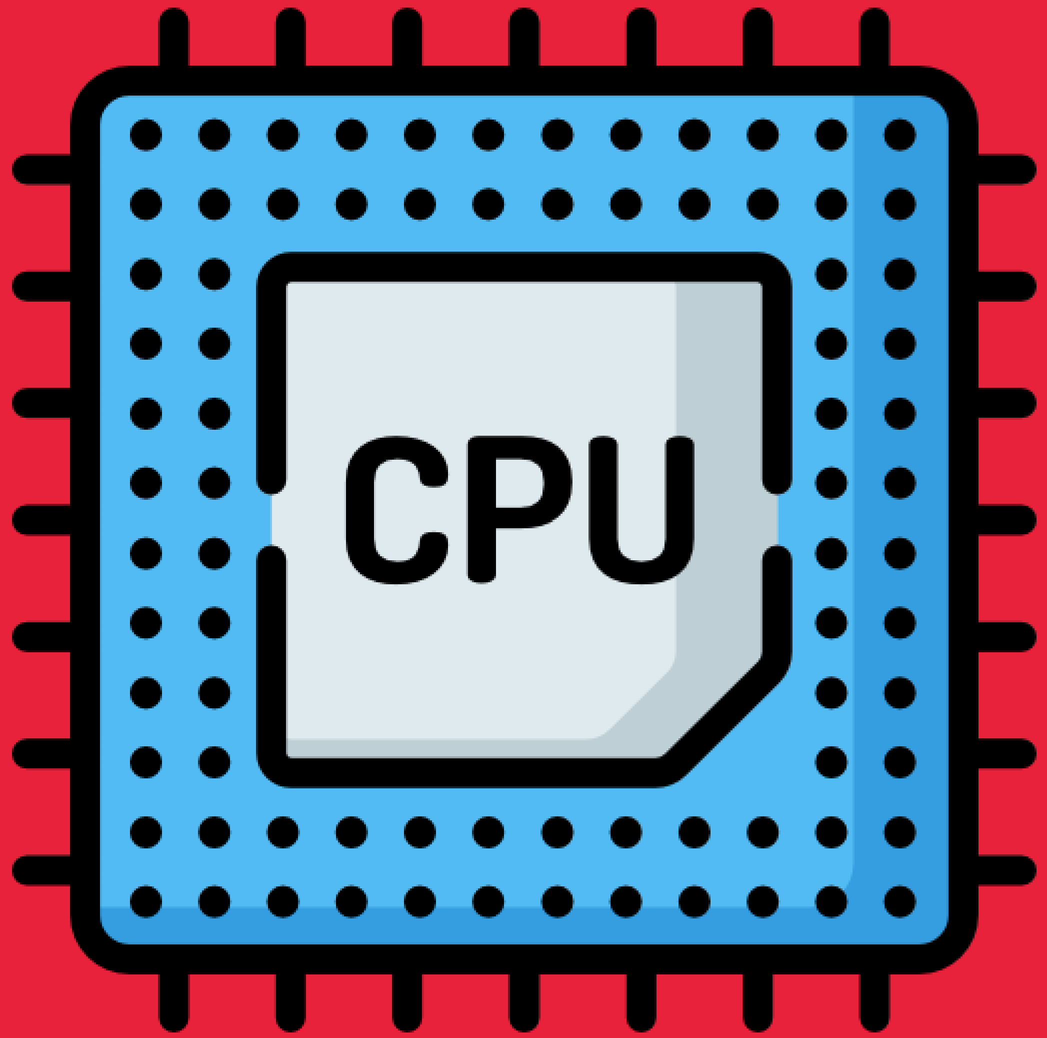


Throughput Units and Processing Speed Unit



"FLOPS" (floating-point operations per second) is a unit of measurement for a computer's performance, indicating how many complex floating-point (decimal) calculations it can perform each second

Throughput Units and Processing Speed Unit



Each CPU has a clock speed for determining how quickly it can do computations. This is commonly about 3.5GHz or 3.5 billion operations per second, and will likely stay around that number for the foreseeable future to physical limitation in the durability of silicon.

Graphical processing units (GPUs) are similarly measured to CPUs, as they are both devices focused on computations. However, with the GPU’s different goal of computing graphics comes different implementations. They tend to have many more computational cores (up to 80!), but with much slower clock speeds (around 2GHz).

Data Types

A data type is a classification of data which tells the compiler or interpreter how the programmer intends to use the data. Most programming languages support various types of data, including integer, real, character or string, and Boolean.

DATA TYPES	REPRESENTS	EXAMPLES
Integer	Whole numbers	-5, 0, 123
Floating points (real)	Fractional numbers	-87.5, 0.0, 3.14159
String	A sequence of characters	“Hello World!”
Boolean	Logical true or false	True, False
Nothing	No Data	Null

Data Representation : ASCII

Description	Character	Hexadecimal Code	ASCII Value
space		20	32
exclamation	!	21	33
Addition sign	+	2B	43
comma	,	2C	44
Hyphen (subtraction)	—	2D	45

- ✓ It is a character encoding standard for electronic communication. American Standard Code for Information Interchange(ASCII) and was first launched in 1963. ASCII codes are used to represent text in computers and telecom devices.
- ✓ ASCII is used for representing 128 English characters in the form of numbers, with each letter being assigned to a specific number in the range 0 to 127. For e.g., the ASCII code for uppercase A is 65, uppercase B is 66, and so on. Check out the following table for some more examples.
- ✓ Most computers are using ASCII encoding for text representation, which makes transferring data from one device to another a lot easier.

Data Representation : Unicode

- ✓ Unicode provides a unique way to define every character in every spoken language of the world by assigning it a unique number. The Unicode standard is maintained by the Unicode Consortium and defines more than 1,40,000 characters from more than 150 modern and historic scripts along with emoji.
- ✓ Unicode can be defined with different character encoding like UTF-8, UTF-16, UTF-32, etc. Among these UTF-8 is the most popular as it used in over 90% of websites on the World Wide Web as well as on most modern Operating systems like Windows.

https://symbl.cc/en/unicode-table/

Data Representation : ASCII vs. Unicode

Key factor-1 :
Size -

1. It is obvious by now that Unicode represents far more characters than ASCII. ASCII uses a 7-bit range to encode just 128 distinct characters. Unicode on the other hand encodes 154 written scripts. And did I mention emoji? Those too.
2. So, we can say that, while Unicode supports a larger range of characters it also takes up a lot more space than ASCII.

Data Representation : ASCII vs. Unicode

Key factor-2 :
ASCII == UNICODE?

1. For backward compatibility, the first 128 Unicode characters point to ASCII characters. And since UTF-8 encodes each of those characters using 1-byte.
2. ASCII is essentially just UTF-8, or we can say that ASCII is a subset of Unicode. Vice versa isn't true.